ELECTRIC VEHICLES | TRANSPORTATION AND MOBILITY

EV Highway Corridor Reliability Gap Analysis



A Python-Based Data Analysis and geospatial analysis of fast-charging coverage on six Indian national highways

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Industry Overview



EV and Mobility Industry

- India's road transport is shifting towards electric vehicles
- 10,377,820 EV registrations in the 2017–2026 dataset (2026 partial)
- Charging networks must keep pace with vehicles on the road



Why Highway Charging Matters

- Long-distance trips depend on fast chargers along the route
- One long uncovered stretch can break an intercity journey
- Highways link states, so gaps are not only a local problem



Role of Data Analysis

- Combines registrations, charger records and road geometry
- Turns coverage into measurable kilometres
- Helps planners decide where new chargers are needed first

Problem Statement

The Challenge

- Fast-charging infrastructure is distributed unevenly across states and routes
- A highway can have many chargers nearby and still contain long stretches without one
- EV registration growth and charger rollout do not always match

Charger count does not equal coverage

243

fast chargers near NH-66

376 km

still uncovered on the same highway

Who Is Affected



EV owners and drivers



Fleet and bus operators



Charging network operators



Planners and policymakers

Project Objectives

1. Analyze EV registrations across states and Union Territories
2. Examine charging station distribution and operational status
3. Evaluate charging coverage on six national highways
4. Identify uncovered highway stretches and long gaps
5. Compare gap indicators using sensitivity analysis and clustering

Proposed Solution and Analysis Questions

EV registrations

Charging stations

National highways

State and UT boundaries

combined into one analysis

Approach

1. Clean and standardise all datasets
2. Split highways into ~25 km segments
3. Draw a 5 km corridor buffer; match fast chargers
4. Flag segments with no fast charger within 50 km
5. Compare states: EV Desert Index, sensitivity, K-Means

Analysis Questions

- Which states and UTs have the most EV registrations?
- How many chargers are operational, and how many are fast (≥25 kW)?
- What share of each of the six highways has fast-charger coverage?
- Where are the longest uncovered stretches (>50 km)?
- Do charger counts alone reflect corridor coverage?
- Which states hold the most uncovered highway length?

Dataset Information

EV Registration Data

- **Source:** Vahan Dashboard
- Year, State/UT, Vehicle category, Registration status
- Format: CSV



36 raw records

EV Charging Station Data

- **Source:** Open Charge Map API
- Geographic coordinates, Operational status, Maximum charging power
- Format: CSV



1,979 raw records

National Highway Data

- **Source:** MoRTH via India Geodata
- Highway attributes and Geographic geometry
- Format: Parquet / GeoJSON



10,317 raw records

State and UT Boundaries

- **Source:** India Geodata
- State/UT names and Polygon Geometry
- Format: Parquet



36 raw records

Key Fields (Combined Dataset)

road_name, state_ut, status, geometry (highways); latitude, longitude, power kW, status (chargers); State, 2017 to 2026 (Vahan); STNAME, geometry (boundaries).

Data Collection and Understanding

Collection

- EV registrations: exported from the Vahan dashboard (Electric, Active, calendar years 2017 to 2026)
- Chargers: Open Charge Map API (free key)
- Highways and boundaries: India Geodata (open licence).
- Boundaries: India Geodata state and UT polygons

The datasets contain a combination of tabular and geospatial information. Registration data supports state- and year-level analysis, while charging stations and highway geometries support spatial distance and coverage analysis.

Inspection checks

Data types

Missing values

Duplicate checks

Geometry validity



Data Cleaning and Transformation

1,979 raw

1,841 cleaned

1,822 operational

Charging station records

1,421

fast chargers (≥25 kW)

10,317 → 9,507

highway records, original → cleaned

Cleaning steps

- Checked and handled duplicate records.
- Checked missing values.
- Validated geographic geometries and coordinates.
- Standardized inconsistent state names.
- Reviewed highway attributes.
- Filtered charging stations based on operational status.
- Prepared cleaned datasets for subsequent analysis.

~25 km highway segment

5 km corridor buffer

50 km gap threshold

≥25 kW fast charger



Exploratory Data Analysis

10,377,820

EV registrations in the 2017–2026 dataset

2026 data is partial

60%

Approximate share of mapped road network identified as uncovered

~98,551 km

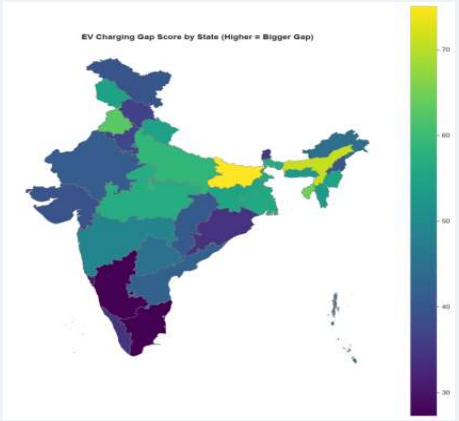
(about 60%) of the mapped road network has no fast charger within 50 km

Analyses performed

- State-level EV registration analysis
- Charger distribution across states
- Highway coverage on six national highways
- Ranked states using the EV Desert Index.
- Sensitivity analysis of gap indicators
- Three-cluster K-Means analysis



Data Visualization

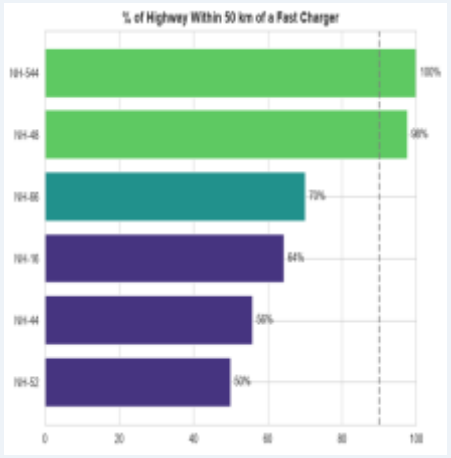


Gap Map: 6 Highways

Purpose: Locate uncovered stretches

Shows: Highways with 50 km gaps marked

Insight: Longest gap: ~427 km on NH-44

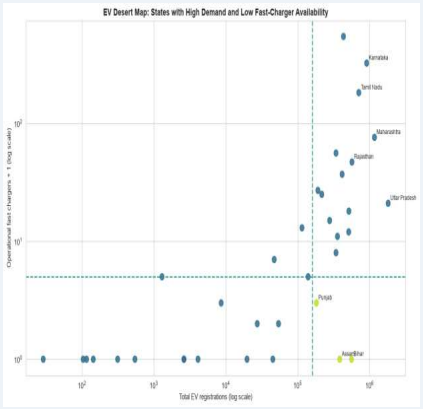


Highway Scorecard

Purpose: Compare highway coverage

Shows: Coverage share per highway

Insight: NH-544 100%, NH-48 97.7%, NH-52 50.0%



EV Desert Quadrant

Purpose: Compare demand and supply

Shows: States by EV registrations vs chargers

Insight: confirm from the figure



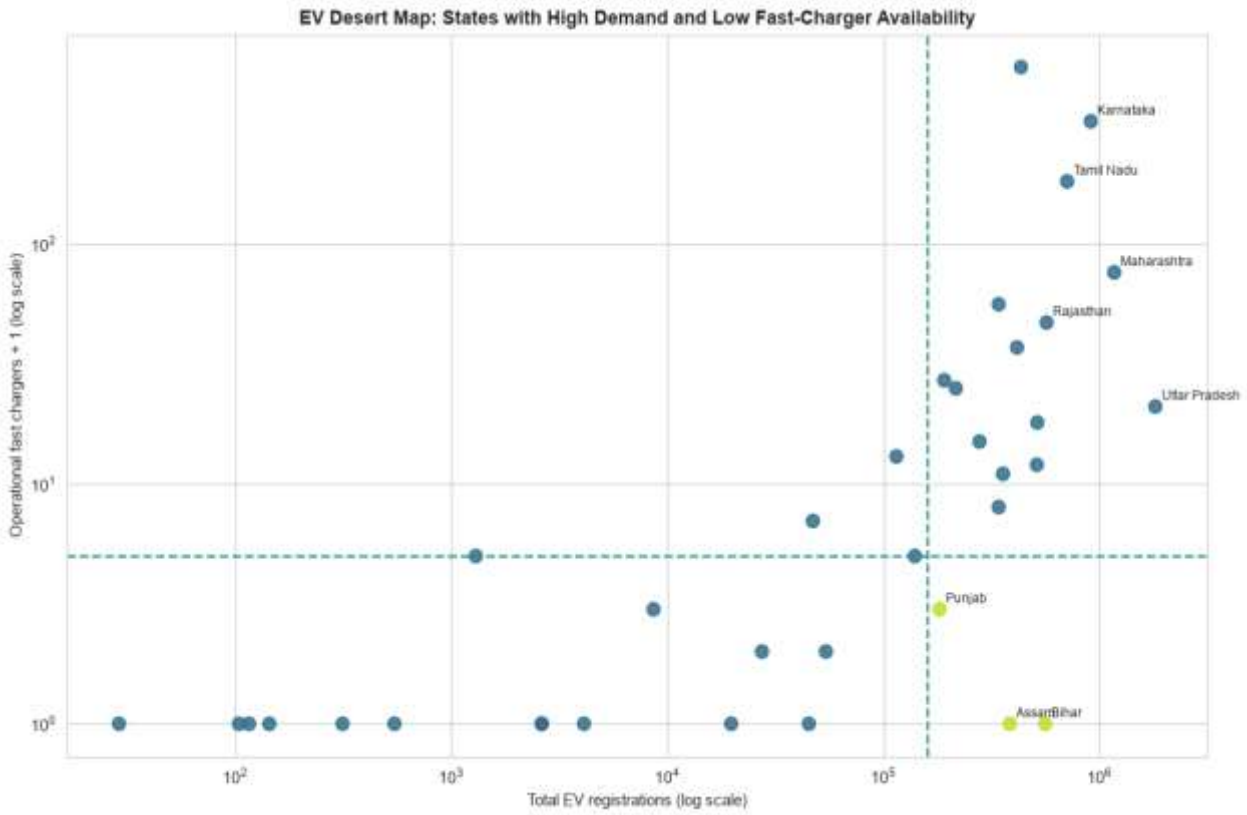
Gap Score Map

Purpose: Map relative gap severity

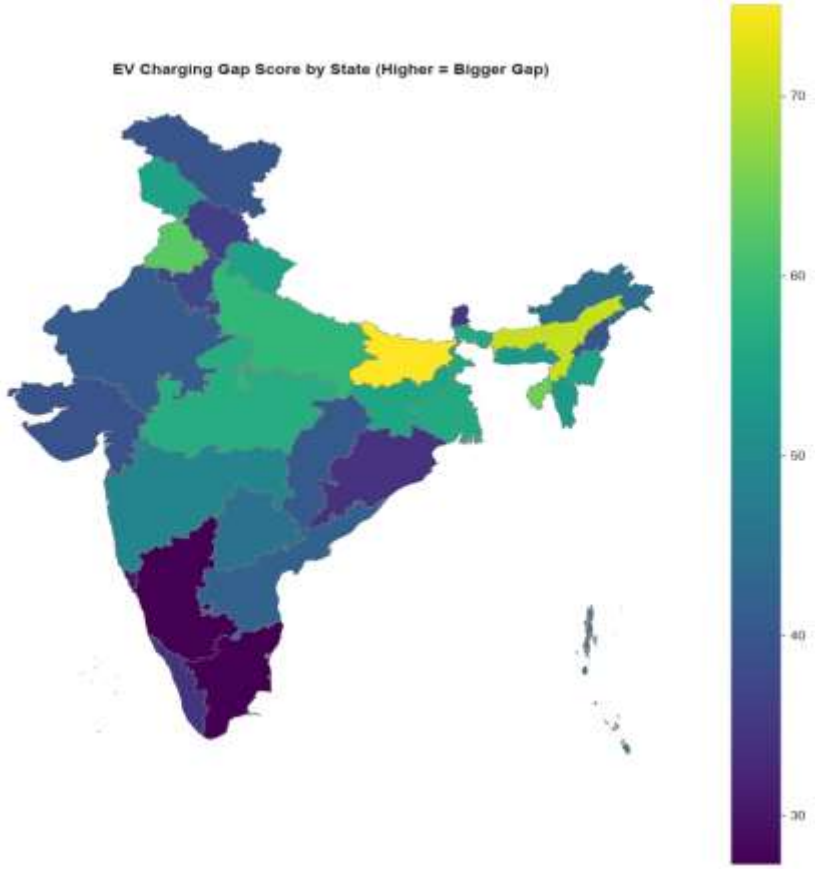
Shows: State-level gap score

Insight: confirm from the figure

Data Visualization

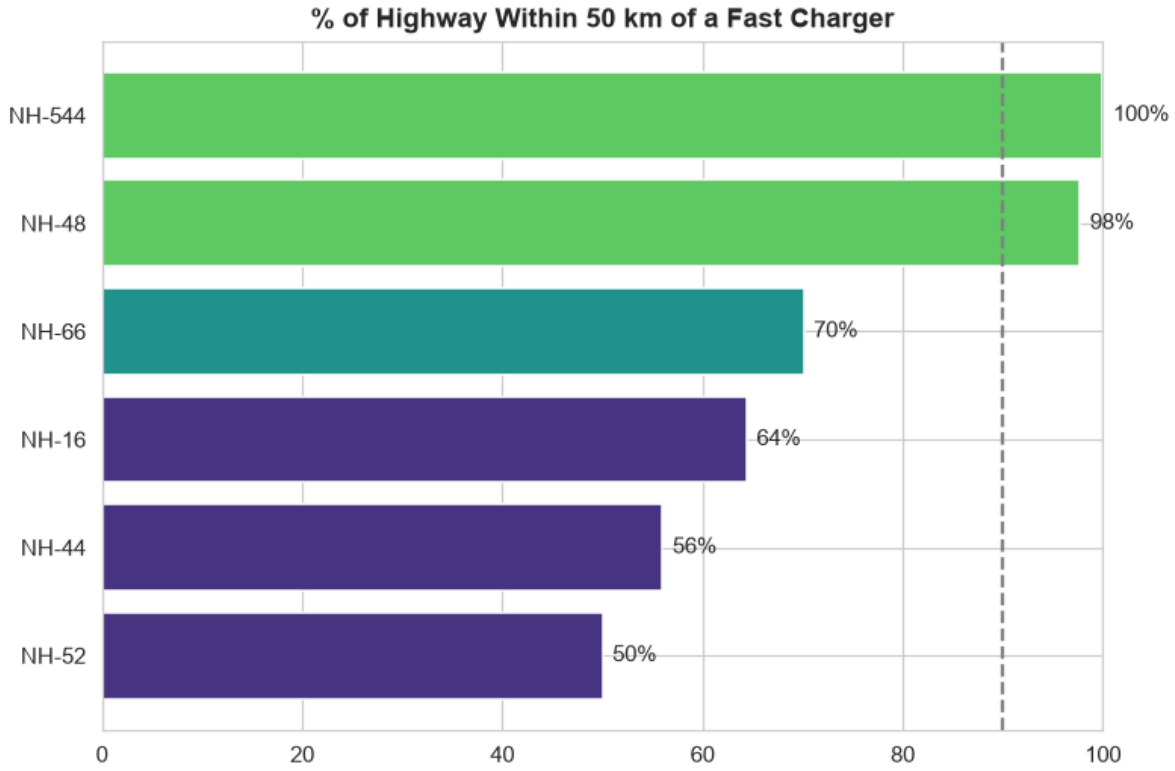


Identifies states with high EV registrations but low operational fast-charger availability. It highlights potential mismatches between EV demand and charging infrastructure.

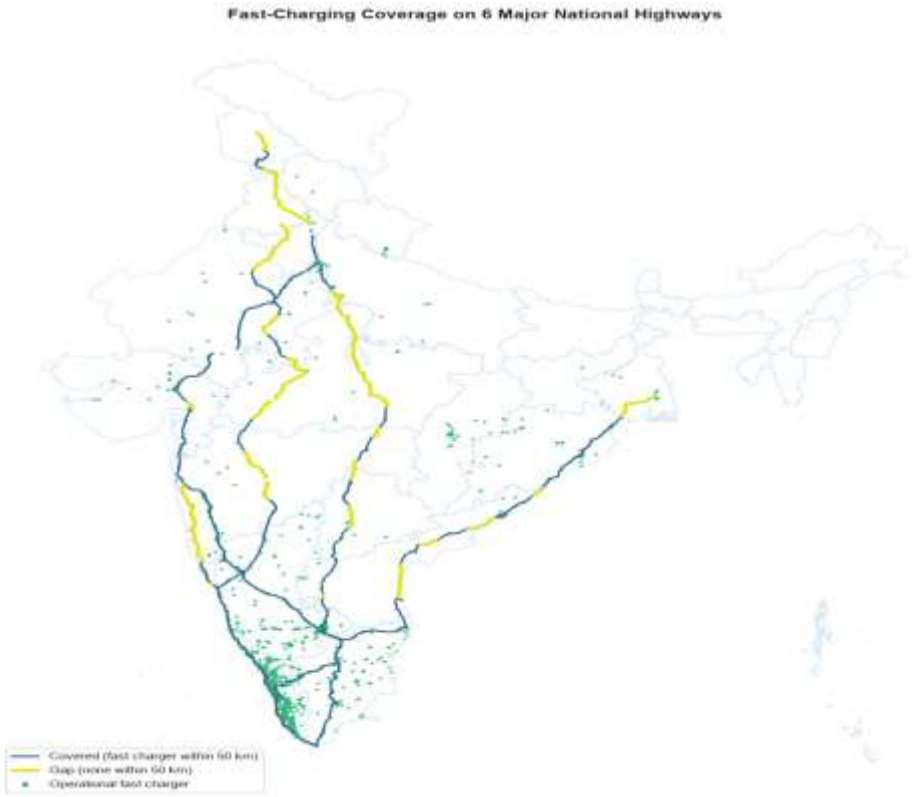


Shows the distribution of EV charging infrastructure gaps across Indian states. Higher scores indicate larger gaps and help identify areas that may need more charging infrastructure.

Data Visualization

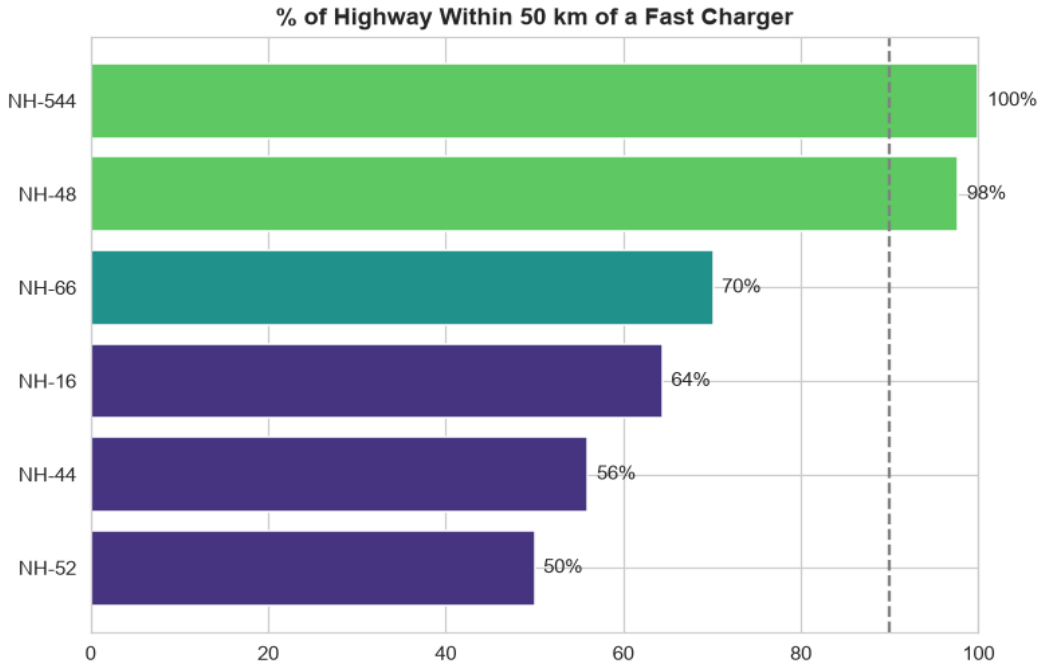


The bar chart compares highway coverage within 50 km of a fast charger, with NH-544 leading at 100% and NH-52 lowest at 50%, highlighting priority areas for infrastructure expansion.



The map highlights fast-charging coverage gaps across six major national highways in India, showing covered routes, uncovered segments, and operational fast chargers.

Key Insights and Recommendations



427 km

longest uncovered stretch (NH-44)

243 vs 376 km

NH-66: nearby fast chargers vs uncovered stretch

~891 km

Maharashtra: uncovered selected-highway length

~796 km

Madhya Pradesh: second highest

Recommendations

Prioritise long uncovered stretches

Validate charger information

Plan for corridor coverage, not charger counts alone

Conclusion and Future Scope

Achievements

- Combined EV registrations, chargers, highways and boundaries in Python
- Cleaned 1,979 charger records to 1,822 operational, 1,421 fast
- Measured coverage on six highways: ~60% of the mapped network has no fast charger within 50 km
- Geographic distribution matters as much as charger counts

Future Scope

- Road-network driving distances
- Traffic data
- Additional highways
- Updated charger information
- Interactive mapping

From counting chargers to understanding charging accessibility — data analysis can help make EV highway infrastructure planning more evidence-based.

Thank you

Any Questions?